2026

$2 + 0 + 2 + 6 = 10 \to 1$

so 2026 leaves remainder 1 when divided by 9

Numbers That Talk

Divisibility shortcuts, and the modular arithmetic hiding behind them

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Is 2026 divisible by 9?

Magic trick 1: the vanishing digit

1. Think of a 4-digit number (not all digits the same).
2. Shuffle its digits into a new number.
3. Subtract: bigger minus smaller.
4. Circle one non-zero digit of the result. Hide it.
5. Read me the other digits, in any order.

example

5274 → **7425**

7425 − 5274 = **2 1 5 1**

They read: 2, 1, 1 → sum 4

4 needs 5 to reach 9. The hidden digit is 5.

If the sum is already a multiple of 9, the answer is 9.

Write along in box 1 of your worksheet

A clock with only 9 marks



walk 10 steps = walk 1 step

$$2026 = 2\cdot 10^3 + 0\cdot 10^2 + 2\cdot 10 + 6$$

on the 9-clock every 10^k is just 1

$$2026 \equiv 2 + 0 + 2 + 6 = 10 \equiv 1 \pmod 9$$

Same digits $\Longrightarrow$ same spot on the clock.

Subtract $\Longrightarrow$ land on the top, that is 0 (mod 9).

So the digits of the result add to a multiple of 9. You read all but one; I fill in the gap.

Same idea, bigger chunks

$$1000 = 999 + 1 \implies 1000 \equiv 1 \pmod{999}, \quad 999 = 27 \times 37$$

So cut a number into three-digit chunks from the right and add them. That tests 27 and 37.

45658 → **45** + **658** = 703 = 19 × 37 ✓

37 divides 45658, no long division. (It is 37 × 1234. We checked.)

Always chunk from the right: 45 | 658, not 456 | 58

Family one: $10^k \equiv 1$

$10 \equiv 1 \pmod 9$ → digit sum

$100 \equiv 1 \pmod{99}$ → 2-digit chunks

$1000 \equiv 1 \pmod{999 = 27 \times 37}$ → 3-digit chunks

7 ? 13 ?

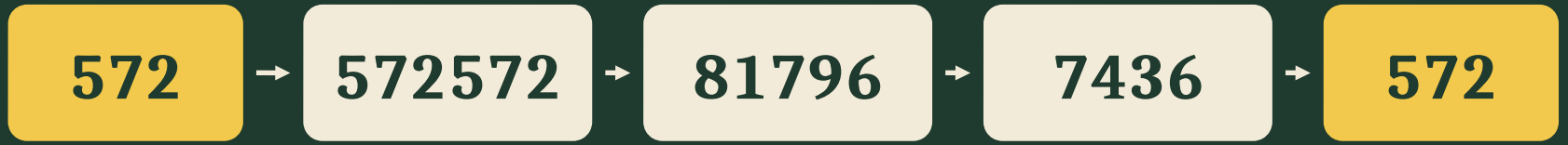
999 = 27 × 37. No 7, no 13 anywhere.

Does 7 even have a rule?

What if $10^k \equiv -1$ instead?

over to Yueran

Magic trick 2: 1001 brings it back



Every division comes out exact. The last one gives back the number you started with. Try your own number.

box 2 of your worksheet

_____ → _____ → ÷ 7 → ÷ 11 → ÷ 13 → _____

Why it works: 1000 acts like −1

$$abcabc = abc \cdot 1000 + abc = abc \cdot 1001 = abc \cdot 7 \cdot 11 \cdot 13$$

And $1000 = 1001 - 1$, so $1000 \equiv -1 \pmod{1001}$.

Cut into three-digit chunks from the right and alternate + − + −. You get the remainder mod 7, 11 and 13 at once.

2026 → **026** − **2** = 24

$24 \equiv 3 \pmod{7}$

$24 \equiv 11 \pmod{13}$

Both are the true remainders of 2026.

572572 → 572 − 572 = 0, so 7, 11 and 13 all divide it. That's the trick again.

One idea, every prime

find k with $10^k \equiv +1$ or $-1 \pmod d$

+1: add k-digit chunks. 9 (k = 1), 37 (k = 3)

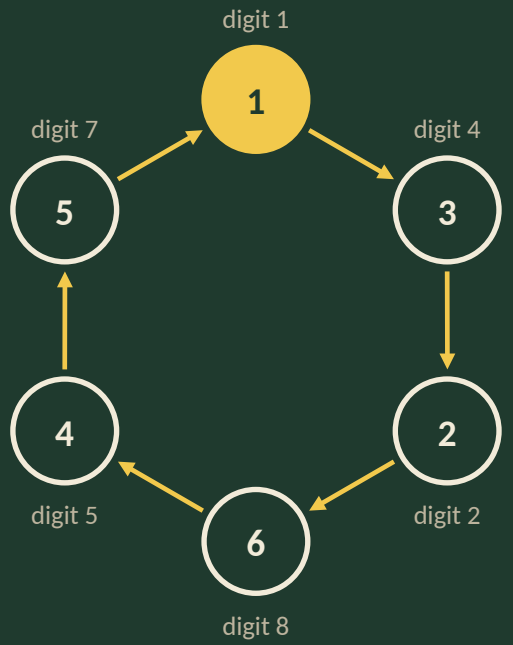
−1: alternate them. 7 and 13 (k = 3)

Fermat: $10^{p-1} \equiv 1 \pmod p$ for every prime $p \neq 2, 5$

So a rule always exists.

Shortest additive chunk = $\text{ord}_p(10)$ = the period of 1/p.

$1/7 = 0.142857\ 142857\ \ldots$ $\quad \text{ord}_7(10) = 6$



remainders of 1 ÷ 7 = powers of 10 mod 7

142857 × 2 = 285714, × 3 = 428571

If you know what 10 is mod d, you can build the rule for d yourself.

Fermat says the rule exists. The order of 10 says how short it can be.

The proofs are on the back of your worksheet.

A quick rule for 7, why ISBN uses 11, 142857 and more: github.com/ZhihengLi0/math5248

Thank you. Now go try the 1001 trick on a friend.